Muhammad Noman

Electrical Engineer | Embedded System

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Professional Summary

Passionate Electrical Engineering student with experience in digital logic design, embedded systems, signal processing, and processor architecture. Proficient in MATLAB and Simulink, Python, SQL, ETAP, and Assembly Language, with hands-on experience in robotics and hardware-based projects. Seeking opportunities to apply my technical knowledge to real-world engineering challenges.

Education

BS Electrical Engineering October 2022 – Present
Namal University, Mianwali, Pakistan

Final Year Project: Design and Development of an Integrated Multiple Sensors Based Inertial Measurement System
Developing a Virtual IMU system of eight redundant IMUs that provides reliable inertial and orientation data despite sensor noise, bias, drift, and failures, using simulation first and then a microcontroller-based hardware platform.

Coursework: Computer Architecture, Digital Logic Design, Digital Signal Processing, Database Engineering, Embedded Systems, Machine Learning, Wireless Communication, Signals and Systems, Power Electronics, Control Systems

Technical Skills

STM32, Embedded Systems, IMU Sensor Interfacing, Assembly Language, Verilog, Digital Logic Design, Proteus, OrCAD Capture, MATLAB, Simulink, Python, SQL, ETAP, AutoCAD, Cisco Packet Tracer, MS Word, MS PowerPoint, MS Excel, Report Making, Documentation, Data Entry, Database Query

Projects

Motion Angle Detection Using 8 IMUs, STM32 and Simulink Embedded Systems

Combined the measurements of eight IMU sensors acquired through an STM32 microcontroller and processed them in MATLAB and Simulink to detect motion angles. The project applied multi-sensor data combination to obtain reliable angle information.

Voice Recognition Using MATLAB Signal Processing

Used nine voice recordings of up to five seconds each, one of which was my own voice. Merged all the recordings into a single audio signal and used MATLAB to recognize and extract my own voice from the combined audio.

Echo Cancellation Using MATLAB Signal Processing

Developed an echo cancellation system in MATLAB to remove echo from an audio signal, and tested the system in MATLAB.

LEGO Mindstorms EV3 Line Following Robot Robotics

Assembled a LEGO Mindstorms EV3 robot and developed a program that enables it to follow a black line. Implemented an open-loop control strategy for the robot's movement using MATLAB and Simulink.

RISC-V Pipelined Processor Design in Verilog Computer Architecture

Designed a pipelined RISC-V processor using the Verilog hardware description language as a course project for Computer Architecture, applying pipelining concepts to processor design.

ALU Design Digital Logic Design

Designed an Arithmetic Logic Unit (ALU), the part of a processor that performs arithmetic and logic operations, as a course project for Digital Logic Design.

Sumo War Robot Namal Tech Expo 2024

Built a sumo war robot and took part in the sumo robot competition at Namal Tech Expo 2024, where I received a participation certificate.

Line Following Robot Namal Tech Expo

Built a line following robot designed to follow a line path and presented it at the Namal Tech Expo.

Bank Networking Design Using Cisco Packet Tracer Computer Communication Networks

Designed a bank network in Cisco Packet Tracer as a course project for Computer Communication Networks, setting up the network devices and connections in the simulator.

Experience

Intern | Northern Power Generation Company Limited (GENCO-III), Thermal Power Station Muzaffargarh July 2025 – October 2025

- Gained practical knowledge of thermal power plant operations and electricity generation.
- Learned the working principles of high-voltage switchyards, switchgear, and circuit breakers.
- Studied the operation, testing, and maintenance of AC and DC motors.
- Gained exposure to industrial control systems, including PLC and SCADA.

Certificates

Internship in Electrical Department, Thermal Power Station (TPS) Muzaffargarh September 2024

Sumo War Robot (Participation), Namal Tech Expo 2024 January 2024

Interests and Languages

Interests: Cricket, Embedded Systems, Problem Solving
Languages: English (Fluent), Urdu (Fluent)